Deliverable 1: Data Model

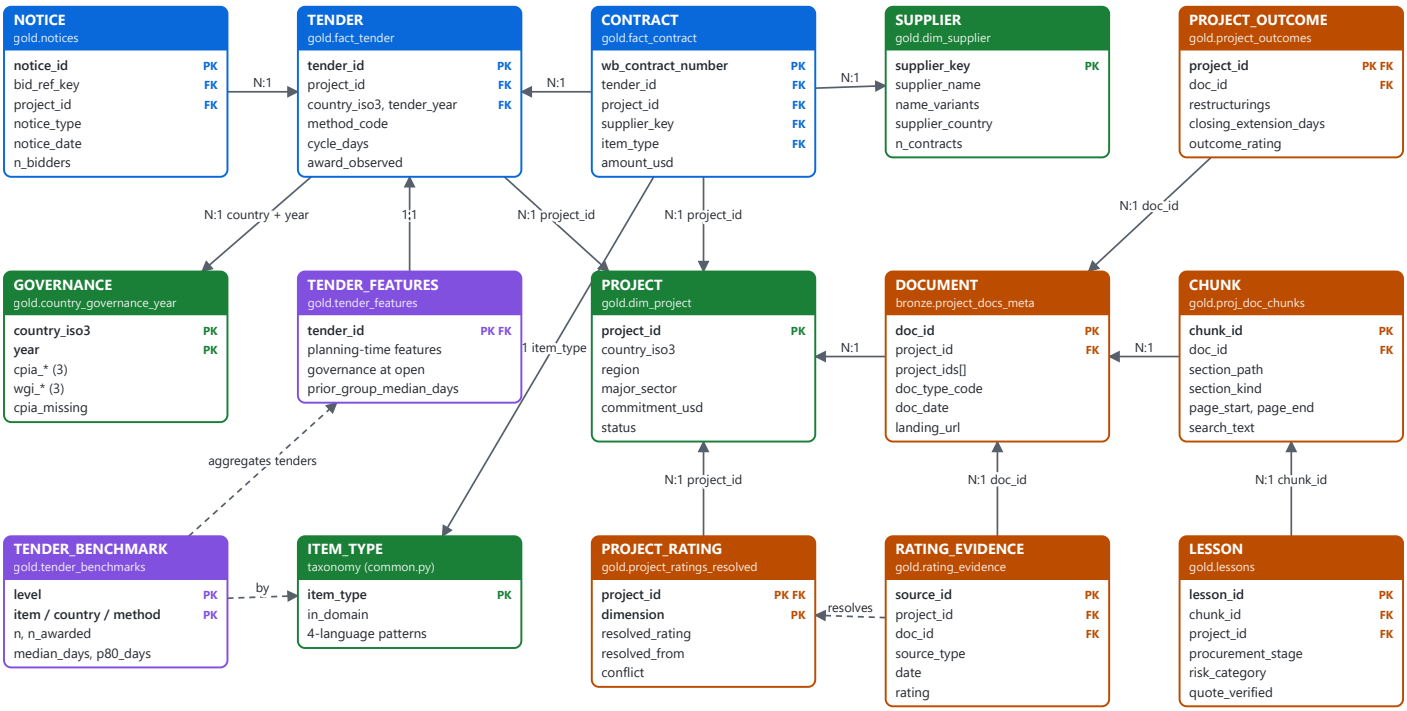
Procurement Intelligence Platform: how structured, semi-structured and unstructured World Bank data fit together

Item	Detail
Physical platform	Databricks Delta tables in Unity Catalog, catalog `proc_demo`: `bronze`, `silver`, `gold`, `ai`, `ops` schemas
Try it	chatbot: https://dmlaishu.github.io/procurement-intelligence/chat/
Domain	Medical laboratory and diagnostic equipment in World Bank-financed health projects
Verified on	local prototype run on public World Bank data, 26 October 2026 (row counts below are from that run); the same tables were built on Databricks on 2026-10-27 (§10)
Companion files	`er_diagram.svg` (diagram), `01_DATA_MODEL_DDL.sql` (full Gold DDL with keys and column comments)

1. Design principles

1. **One conformed spine of keys.** Every source is reduced to a small set of shared identifiers: **Project ID** (P-code), **tender_id** (normalised bid reference), **supplier_key** (resolved supplier), **item_type** (equipment type), **doc_id / chunk_id** and **country_iso3 + year** (with region). Every question that crosses silos crosses through these keys. The hierarchy users navigate is Region → Country → Project → Tender → Contract → Supplier → Equipment type, with documents, passages, lessons and ratings attached to the project.
2. **A star schema for "what happened", a document hierarchy for "why".** Tenders and contracts are facts. Projects, suppliers and country-year governance are dimensions. Documents go *document → chunk → lesson / rating evidence*, so every lesson carries its page and section for citation.
3. **Links are data, with provenance.** Joins that are not exact (fuzzy supplier names, description-matched contracts, LLM-extracted lessons) are stored as rows in a **knowledge-graph edge table** with `method` and `confidence`. A user, a model or the verifier can therefore always ask *how sure is this link?*
4. **Time-aware.** Every fact carries the date it became known. Governance scores are forward-filled, never back-filled, and features use only information available at invitation time. This prevents leakage into models.
5. **Missing is explicit.** Open tenders are kept as censored (`award_observed = false`), unknown outcomes are flagged (`outcome_unknown`), and missing CPIA is a flag (`cpia_missing`), not a zero.

2. Entity-relationship diagram



KG_EDGE (gold.kg_edges) 115,327 edges: one row per link between any two entities above

(src_type, src_id, rel, dst_type, dst_id, method, confidence, valid_from, evidence) method ∈ exact_key | fuzzy_desc | fuzzy_name | temporal | llm_extracted | human_confirmed

FINANCED_BY 14,613 OF_TYPE 14,613 AWARDED_TO 14,608 ANNOUNCES 14,445 FINANCED_BY 10,403 HAS_CONTEXT 10,403 OF_TYPE 10,403 RESULTED_IN 9,782

Facts (what happened) Dimensions Documents and knowledge (how / why) Analytical (model features, benchmarks)

Arrows point from the many side to the one side. Dashed lines are derivations, not foreign keys. `DOCUMENT` is `bronze.project_docs_meta`; in the local prototype the same rows live in `data/raw/documents_meta.json` (131 documents: ISR 36, PP 36, PAD 30, ICR 17, ICRR 12). `ITEM_TYPE` is a reference taxonomy rather than a table: every contract and tender carries its `item_type` and an `in_domain` flag.

3. Layers

Layer	Tables	Role
Bronze	`contract_awards_raw`, `proc_notices_raw`, `projects_raw`, `governance_raw`, `country_ref`, `domain_projects`, `project_docs_meta` + volume `raw_docs`; `rag_documents` + volume `rag_docs` (§7.1)	API records and PDFs exactly as received, with ingest timestamp; Change Data Feed on
Silver	`contracts`, `notices`, `projects`, `governance`, `tenders`, `proj_doc_pages`, `rag_pages`	typed, de-duplicated, keys normalised (ISO codes, bid references, supplier names); one page of text per row
Gold	facts, dimensions, documents and knowledge, ML tables, `kg_edges` (below), view `v_lab_contracts`, `rag_chunks` (§7.1)	resolved entities and links; what the models, the vector indexes, the copilot and the chatbot read
AI	UC function tools (`proc_*`), vector indexes `proj_doc_chunks_index` and `rag_chunks_index`, registered models `proc_*`	what the copilot and chatbot call; governed by Unity Catalog grants
Ops	`app_logs`, `proc_briefs`, `proc_feedback`, `v_proc_kpis`, `rag_countries`, `rag_eval_results`	run log, every brief generated, user feedback, realised award dates, the country registry of the document Q&A and its evaluation history

4. Table catalogue

Table	Layer	Grain (one row per)	Primary key	Foreign keys / joins	Rows (prototype)	Source
`notices`	Gold (silver.notices)	one procurement notice	`notice_id`	bid_ref_key → fact_tender; project_id → dim_project	15,158	Procurement Notices API
`fact_tender`	Gold	one tender (bid reference), from invitation to award	`tender_id`	project_id; (country_iso3, tender_year) → governance; supplier_key	10,403	built from notices
`fact_contract`	Gold	one contract-supplier row	`wb_contract_number + supplier_key`	tender_id; project_id; supplier_key	14,613	Major Contract Awards (FinancesOne)
`dim_supplier`	Gold	one resolved supplier	`supplier_key`	-	7,540	resolved from contract supplier names
`dim_project`	Gold	one project	`project_id`	country_iso3	179	Projects API
`country_governance_year`	Gold	country × year	`(country_iso3, year)`	-	1,290	WDI API: CPIA + WGI
`project_docs_meta`	Bronze	one project document	`doc_id`	project_id (project_ids[] for PPARs)	131	Documents & Reports API
`proj_doc_pages`	Silver	document page	`(doc_id, page_num)`	doc_id	(Databricks only)	PDF text
`proj_doc_chunks`	Gold	section-bounded passage (≤ 450 words)	`chunk_id`	doc_id; project_id	6,187	parsed PDFs
`lessons`	Gold	one lesson learned	`lesson_id`	chunk_id; doc_id; project_id	150	Lessons / Key factors sections
`rating_evidence`	Gold	one dated DO / IP rating	`source_id + dimension`	project_id; doc_id	301	ICR data sheets, ISRs, PPARs
`project_ratings_resolved`	Gold	project × rating dimension	`(project_id, dimension)`	project_id	34	rating_evidence
`project_outcomes`	Gold	one completed project	`project_id`	doc_id	17	ICR data sheet
`tender_features`	Gold (ML)	one tender with ≥ invitation date	`tender_id`	tender_id	1,632	fact_tender + project + governance

Table	Layer	Grain (one row per)	Primary key	Foreign keys / joins	Rows (prototype)	Source
`tender_benchmarks`	Gold (ML)	benchmark cohort	`(level, country, category, method)`	-	67	Kaplan-Meier over fact_tender
`item_types`	Reference	one equipment type	`item_type`	-	14	taxonomy in procurement_demo/common.py
`kg_edges`	Gold	one typed link	`(src_type, src_id, rel, dst_type, dst_id)`	any entity ↔ any entity	115,327	all of the above
`proc_briefs`	Ops	one generated brief	`brief_id`	evidence_ids[] → any entity	(Databricks only)	copilot
`proc_feedback`	Ops	feedback on a brief / evidence item	`(brief_id, evidence_id, created_ts)`	brief_id	(Databricks only)	users; realised award dates

5. How disjoint keys are reconciled

Link	Rule	Edge `method` / confidence	Prototype result
Notice → Tender	normalised bid reference (`KE-MOWASCO-16771-CW-RFB` ≡ `KEMOWASCO16771CWRFB`)	`exact_key` / 1.0	10,403 tenders
Contract → Tender	(1) normalised borrower contract reference = bid reference; (2) else same project + category, signed ≤ 18 months after invitation, description token Jaccard ≥ 0.5	`exact_key` 1.0 / `fuzzy_desc` = similarity	9,600 exact + 182 fuzzy = **67%** of contracts
Contract → Supplier	normalised name key (drops M/S, legal forms, punctuation); within blocks of country + first token, similarity ≥ 0.92 → union-find merge	`exact_key` / `fuzzy_name`	7,835 raw names → **7,540** suppliers (94 fuzzy merges)
Contract / Tender → Equipment type	description classified by an ordered taxonomy of patterns in English, French, Spanish and Portuguese (tolerant of garbled accents); generic classes are blocked for vehicles, CCTV and printing	`keyword` / 0.7	2,220 of 14,613 contracts are laboratory or diagnostic equipment
Contract / Tender / Document → Project	Project ID; PPARs covering several projects fan out to each	`exact_key`	**100%** of contracts
Tender → Governance	country + year of invitation, forward-filled ≤ 3 years	`temporal`	all tenders
Lesson → Chunk → Document → Project	lesson keeps its chunk and page; chunk keeps its document	`rule_numbered_paragraph` (local) / `llm_extracted` (Databricks)	150 lessons, 147 quote-verified
Country identifiers	ISO-2 ↔ ISO-3 ↔ names in notices (incl. "Viet Nam" / "Vietnam")	alias table	-

Matches with confidence between 0.6 and 0.9 go to a review queue. Confirmed decisions are written back as `method = 'human_confirmed'` and reused.

6. Knowledge graph (`gold.kg_edges`)

The relational tables answer known questions fast. The edge table answers *path* questions: "which lessons come from projects where this supplier won Works contracts?". It is a property graph stored as one Delta table, clustered by `(rel, dst_id)` for reverse lookups. No separate graph database is needed at this scale.

From → To	Relationship	Edges
CONTRACT → PROJECT	`FINANCED_BY`	14,613
CONTRACT → ITEM_TYPE	`OF_TYPE`	14,613
CONTRACT → SUPPLIER	`AWARDED_TO`	14,608
NOTICE → TENDER	`ANNOUNCES`	14,445
TENDER → PROJECT	`FINANCED_BY`	10,403
TENDER → GOVERNANCE	`HAS_CONTEXT`	10,403
TENDER → ITEM_TYPE	`OF_TYPE`	10,403
TENDER → CONTRACT	`RESULTED_IN`	9,782
PROJECT → SECTOR	`IN_SECTOR`	7,540
SUPPLIER → COUNTRY	`BASED_IN`	7,540
PROJECT → COUNTRY	`LOCATED_IN`	179
LESSON → STAGE	`AT_STAGE`	167
LESSON → PROJECT	`ABOUT`	150
LESSON → CHUNK	`FROM`	150
DOCUMENT → PROJECT	`DESCRIBES`	131
TENDER → STATE	`IN_STATE`	109
COUNTRY → REGION	`IN_REGION`	87
PROJECT → STATE	`LOCATED_IN`	2
STATE → COUNTRY	`PART_OF`	2

Total: 115,327 edges. Three-hop example (contract → supplier → project → ICR lesson):

```
SELECT l.lesson, l.evidence_quote, l.title, l.page_start
FROM gold.kg_edges a                                          -- CONTRACT -AWARDED_TO-> SUPPLIER
JOIN gold.kg_edges f ON f.src_id = a.src_id AND f.rel = 'FINANCED_BY'    -- CONTRACT -FINANCED_BY-> PROJECT
JOIN gold.lessons  l ON l.project_id = f.dst_id AND l.quote_verified
JOIN gold.fact_contract c ON c.wb_contract_number = a.src_id AND c.procurement_category = 'Works'
WHERE a.rel = 'AWARDED_TO' AND a.dst_id = :supplier_key;
```

7. Document model (unstructured side)

```
DOCUMENT (PAD / ISR / ICR / PPAR, 60-150 pages)
  └ PAGE            silver.proj_doc_pages     text per page, line breaks kept, running headers removed
     └ CHUNK        gold.proj_doc_chunks      bounded by headings (Roman / letter / ANNEX), ≤ 450 words, 60 overlap
       │                                      section_path, section_kind, content_type, page_start-page_end
       ├ LESSON          gold.lessons             verbatim evidence_quote checked against the chunk
       └ (vector index)  search_text embedded     doc type | project | country | sector | date | section + text
  └ DATA SHEET (ICR) → RATING_EVIDENCE (dated ISR/ICR/PPAR ratings) → PROJECT_RATING (resolved, conflict flag)
                     → PROJECT_OUTCOME (restructurings, closing extension, outcome ratings)
```

Prototype: 131 reports → 6,187 chunks → 150 lessons and 301 dated ratings.

7.1 Document Q&A by country

The document Q&A keeps its own, country-partitioned copy of the document hierarchy, so a country can be added or refreshed without touching the others. It joins the rest of the model through **Project ID** and **country ISO-3**.

```
RAG_COUNTRY (ops.rag_countries)          iso3, iso2, name, pinned projects
  └ RAG_DOCUMENT (bronze.rag_documents)  country + doc_id; doc type, title, date, project_id → dim_project, file in rag_docs/<country>/<type>/
      └ PAGE (silver.rag_pages)          doc_id + page_num
          └ PASSAGE (gold.rag_chunks)    chunk_id = <country>_<doc_id>_<order>; section, pages, text, 1/0 category columns
              └ (ai.rag_chunks_index)    Vector Search on search_text, filtered by country and category
```

Table	Grain (one row per)	Key	Joins	Rows
`ops.rag_countries`	country	`iso3`	-	3 (local), 2 (Databricks)
`bronze.rag_documents`	country × document	`(country, doc_id)`	`project_id` → `dim_project`	111 (local)
`silver.rag_pages`	document page	`(doc_id, page_num)`	`doc_id` → `rag_documents`	-
`gold.rag_chunks`	passage	`chunk_id`	`doc_id` , `country` , `project_id`	4,761 (local), 3,795 (Databricks)
`ai.rag_chunks_index`	passage vector	`chunk_id`	synced from `gold.rag_chunks`	11 category filters + country

Category columns (`cat_*` , 1/0) mark which equipment types a passage mentions (the same taxonomy as `item_type`) and whether it is about the procurement process; the chatbot's category menu filters on them.

8. Data dictionary

Null shares and examples are measured on the prototype tables. High null shares are expected where an event has not happened yet (e.g. `award_date` for open tenders) or a field is not published for older projects.

`gold.dim_project` (179 rows)

Grain: one project.

Column	Type	Key	Null	Meaning	Example
`project_id`	STRING	PK	0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P149340`
`country_iso3`	STRING		13%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	`IND`
`project_name`	STRING		24%	Project name	`Nagaland Health Project`
`approval_date`	DATE		30%	Board approval date	`2016-12-19`
`closing_date`	DATE		24%	Current closing date	`2024-06-30`
`status`	STRING		24%	Active | Closed | Pipeline | Dropped	`Closed`
`lending_instrument`	STRING		24%	IPF, PforR, DPF, ...	`Investment Project Financing`
`commitment_usd`	DOUBLE		24%	Total commitment (USD)	`48000000.0`
`env_category`	STRING		63%	Environmental category A/B/C/F (older projects)	`B`
`es_risk_rating`	STRING		100%	Environmental & social risk rating (ESF projects; NULL before ESF)	
`major_sector`	STRING		24%	Major sector of the project	`Health`
`implementing_agency`	STRING		26%	Implementing agency	`Department of Health and Family Welfare, Governm…`

Column	Type	Key	Null	Meaning	Example
`project_url`	STRING		0%	Project page	`https://projects.worldbank.org/en/projects-opera…`
`objective`	STRING		0%	Project development objective	
`region`	STRING		2%	World Bank region of the borrower country	`South Asia`
`state`	STRING		99%	Sub-national state (India scope only)	`Nagaland`
`n_tenders`	DOUBLE		22%	Tenders observed for the project	`53.0`
`median_cycle_days`	DOUBLE		41%	Median invitation-to-award days for the project's tenders	`208.0`

`gold.dim_supplier` (7,540 rows)

Grain: one resolved supplier.

Column	Type	Key	Null	Meaning	Example
`supplier_key`	STRING	PK	0%	Resolved supplier: normalised name key + supplier country (after fuzzy merge)	`b hler farma|NI`
`supplier_name`	STRING		0%	Supplier name as published	`BÜHLER FARMA, S.A.`
`name_variants`	STRING		0%	JSON list of all raw spellings merged into this supplier	`["B\u00dcHLER FARMA, S.A."]`
`n_contracts`	BIGINT		0%	Contracts awarded to this supplier	`9`
`total_usd`	DOUBLE		0%	Total awarded value (USD)	`1305730.25`
`supplier_country`	STRING		0%	Supplier country	`Nicaragua`

`gold.country_governance_year` (1,290 rows)

Grain: country × year.

Column	Type	Key	Null	Meaning	Example
`country_iso3`	STRING	PK	0%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	`RWA`
`year`	BIGINT	PK	0%	Calendar year of the indicator (forward-filled ≤ 3 years, never back-filled)	`2017`
`cpia_budget_financial_mgmt`	DOUBLE		34%	CPIA quality of budgetary and financial management (1-6)	`4.0`
`cpia_public_sector_mgmt`	DOUBLE		34%	CPIA public sector management and institutions cluster (1-6)	`3.7`
`cpia_transparency_corruption`	DOUBLE		34%	CPIA transparency, accountability and corruption (1-6)	`3.5`
`wgi_control_of_corruption`	DOUBLE		0%	WGI control of corruption (about -2.5 to 2.5)	`0.5040962`
`wgi_gov_effectiveness`	DOUBLE		0%	WGI government effectiveness	`0.3584605`
`wgi_regulatory_quality`	DOUBLE		0%	WGI regulatory quality	`0.1621531`
`cpia_missing`	BIGINT		0%	1 when CPIA is not published (non-IDA countries)	`0`

`gold.notices` (15,158 rows)

Grain: one procurement notice.

Column	Type	Key	Null	Meaning	Example
notice_id	STRING	PK	0%	Procurement notice id (OP...)	OP00241332
country_iso3	STRING		18%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	KEN
notice_type	STRING		0%	Invitation for Bids, Request for Expression of Interest, Contract Award, ...	Contract Award
notice_date	DATE		0%	Publication date of the notice	2023-08-09
deadline_date	DATE		89%	Bid / EOI submission deadline	2023-04-06
project_id	STRING		0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	P152394
bid_ref	STRING		0%	Borrower bid reference as published	KE-HOMABAY COUNTY-269732-GO-RFQ
description	STRING		0%	Free-text description of the package / contract	SUPPLY OF 1NO. ANAESTHETIC MACHINE
procurement_category	STRING		2%	Works | Goods | Consultant Services | Non-consulting Services	Goods
method_code	STRING		0%	Method code: RFB, RFQ, QCBS, CQS, INDV, DIR, ...	RFQ
method_name	STRING		0%	Method as published on the notice	Request for Quotations
implementing_org	STRING		88%	Implementing agency named on the notice	Project Coordination Unit (PCU)
rebid_text	BOOLEAN		0%	Notice text indicates a re-tender / re-advertisement	False
award_date	DATE		62%	Earliest contract-award notice date	2021-05-20
n_bidders	DOUBLE		12%	Number of bids / proposals received (award notice)	3.0
n_rejected	DOUBLE		12%	Number of bids rejected (award notice)	0.0
rejection_reasons	STRING		12%	Reasons for rejection, as published	
awarded_names	STRING		12%	Awarded firm(s) named on the award notice	PHISSIAL ENTERPRISES
contract_duration_months	DOUBLE		13%	Contract duration on the award notice	3.0
notice_version	DOUBLE		13%	Amendment counter of the award notice	1.0
bid_ref_key	STRING		3%	Normalised bid reference; equals tender_id	KEHOMABAYCOUNTY269732GORFQ

`gold.fact_tender` (10,403 rows)

Grain: one tender (bid reference), from invitation to award.

Column	Type	Key	Null	Meaning	Example
tender_id	STRING	PK	0%	Normalised bid reference (upper-case, punctuation removed); the tender's identity	JKMSCLOTAUTOCLAVESINGLEBIN09
open_date	DATE		84%	First invitation (IFB / REOI) date: start of the procurement cycle	2020-12-23
deadline_date	DATE		85%	Bid / EOI submission deadline	2021-01-09
n_invitations	BIGINT		0%	Number of invitation notices for the same bid reference (re-advertisements)	1

Column	Type	Key	Null	Meaning	Example
`project_id`	STRING	FK	0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P154990`
`country_iso3`	STRING		19%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	`IND`
`procurement_category`	STRING		1%	Works | Goods | Consultant Services | Non-consulting Services	`Goods`
`procurement_method`	STRING		0%	Full method name (Request for Bids, QCBS, ...)	`Request for Bids`
`method_code`	STRING		0%	Method code: RFB, RFQ, QCBS, CQS, INDV, DIR, ...	`RFB`
`description`	STRING		0%	Free-text description of the package / contract	`Procurement of 100 No. Autoclave Vertical Single…`
`implementing_org`	STRING		84%	Implementing agency named on the notice	`Jammu and Kashmir Medical Supplies Corporation L…`
`open_notice_type`	STRING		84%	Type of the opening notice	`Invitation for Bids`
`award_date`	DATE		5%	Earliest contract-award notice date	`2021-06-21`
`n_bidders`	DOUBLE		5%	Number of bids / proposals received (award notice)	`1.0`
`n_rejected`	DOUBLE		5%	Number of bids rejected (award notice)	`0.0`
`rejection_reasons`	STRING		5%	Reasons for rejection, as published	
`awarded_names`	STRING		5%	Awarded firm(s) named on the award notice	`NEW ALPINE TRADERS`
`contract_duration_months`	DOUBLE		7%	Contract duration on the award notice	`2.0`
`rebid_flag`	BOOLEAN		0%	Target: re-tendered (re-advertised or flagged as re-tender)	`False`
`award_amended_flag`	BOOLEAN		0%	Award notice was amended	`False`
`award_observed`	BOOLEAN		0%	An award notice exists (event observed)	`True`
`observed_days`	DOUBLE		90%	Days from open to award, or to the data cut if still open (survival time)	`180.0`
`outcome_unknown`	BOOLEAN		0%	No award after 2 years: excluded as unknown outcome, not censored	`False`
`cycle_days`	DOUBLE		89%	Target: days from invitation to award (awarded tenders only)	`180.0`
`bid_window_days`	DOUBLE		85%	Days between invitation and submission deadline	`17.0`
`single_bid_flag`	DOUBLE		5%	Target: exactly one responsive bid	`1.0`
`rejected_share`	DOUBLE		27%	Share of bids rejected	`0.0`
`tender_year`	DOUBLE		0%	Year of invitation (joins governance by country + year)	`2020.0`
`review_type`	STRING		20%	Prior | Post review by the Bank ('unknown' when not published)	`Post`
`contract_usd`	DOUBLE		20%	Awarded contract value (from the linked contract)	`100792.77`
`supplier_key`	STRING		21%	Resolved supplier: normalised name key + supplier country (after fuzzy merge)	`new alpine traders|IN`

Column	Type	Key	Null	Meaning	Example
`state_from`	STRING		99%	Where the state came from: implementing agency or project	`agency`
`state`	STRING		99%	Sub-national state (India scope only)	`Jammu and Kashmir`
`item_type`	STRING		0%	What was bought, classified into the scope's taxonomy (e.g. PCR, imaging, reagents, analyzers)	`Biosafety, sterilization and cold chain`
`in_domain`	BOOLEAN		0%	Item type belongs to the domain (laboratory and diagnostic equipment)	`True`
`region`	STRING		2%	World Bank region of the borrower country	`South Asia`

`gold.fact_contract` (14,613 rows)

Grain: one contract-supplier row.

Column	Type	Key	Null	Meaning	Example
`wb_contract_number`	STRING	PK	0%	World Bank contract number	`1686395`
`supplier_id`	STRING		0%	WB supplier id (kept as evidence; not trusted alone)	`640252`
`supplier_name`	STRING		0%	Supplier name as published	`RAMIETECH SOLUTIONS`
`supplier_country`	STRING		0%	Supplier country	`India`
`supplier_country_code`	STRING		0%	Supplier country ISO-2	`IN`
`project_id`	STRING	FK	0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P149340`
`project_name`	STRING		0%	Project name	`Nagaland Health Project`
`country_iso3`	STRING		12%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	`IND`
`region`	STRING		1%	World Bank region of the borrower country	`South Asia`
`fiscal_year`	BIGINT		0%	WB fiscal year of the award	`2022`
`signing_date`	DATE		0%	Contract signing date	`2022-01-18`
`procurement_category`	STRING		0%	Works | Goods | Consultant Services | Non-consulting Services	`Goods`
`procurement_method`	STRING		0%	Full method name (Request for Bids, QCBS, ...)	`Request for Bids`
`review_type`	STRING		0%	Prior | Post review by the Bank ('unknown' when not published)	`Post`
`amount_usd`	DOUBLE		0%	Contract amount (USD)	`475879.8`
`description`	STRING		0%	Free-text description of the package / contract	`Supply and Installation of IT equipment and Netw…`
`borrower_ref`	STRING		0%	Borrower contract reference as published	`NHP/PP/2021/A/005`
`borrower_ref_key`	STRING		3%	Normalised borrower reference (exact-key link to tender_id)	`NHPPP2021A005`
`supplier_name_key`	STRING		0%	Normalised supplier name (legal forms, M/S, punctuation removed)	`ramietech solutions`

Column	Type	Key	Null	Meaning	Example
item_type	STRING		0%	What was bought, classified into the scope's taxonomy (e.g. PCR, imaging, reagents, analyzers)	IT, furniture and vehicles
in_domain	BOOLEAN		0%	Item type belongs to the domain (laboratory and diagnostic equipment)	False
supplier_key	STRING	PK	0%	Resolved supplier: normalised name key + supplier country (after fuzzy merge)	ramietech solutions|IN
tender_id	STRING	FK	33%	Normalised bid reference (upper-case, punctuation removed); the tender's identity	NHPPP2021A005
tender_link_method	STRING		33%	How the contract was linked to its tender: exact_key | fuzzy_desc	exact_key
tender_link_confidence	DOUBLE		33%	Confidence of the tender link (1.0 exact; similarity otherwise)	1.0
state	STRING		99%	Sub-national state (India scope only)	Nagaland

gold.proj_doc_chunks (6,187 rows)

Grain: section-bounded passage (≤ 450 words).

Column	Type	Key	Null	Meaning	Example
chunk_id	STRING	PK	0%	_; one section-bounded passage	34459657_0004
doc_id	STRING		0%	Documents & Reports API id (also the PDF file name)	34459657
doc_type_code	STRING		0%	PAD | ISR | ICR | PPAR | PROC_PLAN	ISR
project_id	STRING		0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	P180849
country_iso3	STRING		19%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	TUR
doc_date	STRING		0%	Document date	2025-02-25
title	STRING		0%	Document title (used in citations)	Disclosable Version of the ISR - Türkiye Earthqu…
landing_url	STRING		0%	Document page on documents.worldbank.org (citation link)	http://documents.worldbank.org/curated/en/099022…
section_path	STRING		0%	Heading path of the chunk (e.g. 'V. LESSONS > B. ...')	FRONT MATTER > 4. KEY ISSUES & STATUS
section_kind	STRING		0%	lessons | key_factors | procurement_fiduciary | risk | ratings | results | design | annex | other	other
content_type	STRING		0%	narrative | table | annex	narrative
page_start	BIGINT		0%	First page of the passage (citation)	2
page_end	BIGINT		0%	Last page of the passage	2
n_words	BIGINT		0%	Words in the chunk (≤ 450, 60 overlap)	174
chunk_text	STRING		0%	Passage text, running headers removed	4.1 Implementation Status and Key Decisions The …
search_text	STRING		0%	Contextual text that is embedded: doc type | project | country | sector | date | section + text	ISR | Türkiye Earthquake Recovery and Reconstruc…

`gold.lessons` (150 rows)

Grain: one lesson learned.

Column	Type	Key	Null	Meaning	Example
`lesson_id`	STRING	PK	0%	_L	`34318883_L0071`
`doc_id`	STRING		0%	Documents & Reports API id (also the PDF file name)	`34318883`
`chunk_id`	STRING	FK	0%	_; one section-bounded passage	`34318883_0046`
`project_id`	STRING		0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P157291`
`country_iso3`	STRING		28%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	`KHM`
`doc_type_code`	STRING		0%	PAD | ISR | ICR | PPAR | PROC_PLAN	`ICR`
`section_kind`	STRING		0%	lessons | key_factors | procurement_fiduciary | risk | ratings | results | design | annex | other	`key_factors`
`paragraph`	DOUBLE		2%	Numbered paragraph in the report	`74.0`
`page_start`	BIGINT		0%	First page of the passage (citation)	`27`
`lesson`	STRING		0%	Lesson statement (first sentence)	`Construction of the two PRHs financed by the Pro…`
`detail`	STRING		0%	Full paragraph	`Construction of the two PRHs financed by the Pro…`
`procurement_stage`	STRING		0%	planning | tender_design | evaluation_award | contract_management | general	`planning;contract_management`
`risk_category`	STRING		0%	delay | capacity | fiduciary | safeguards | cost_overrun | specification | other	`delay`
`procurement_related`	BOOLEAN		0%	Lesson concerns procurement	`True`
`evidence_quote`	STRING		0%	Verbatim quote; must be found in the source chunk	`Construction of the two PRHs financed by the Pro…`
`extraction_method`	STRING		0%	rule_numbered_paragraph (local) | llm_extracted (Databricks)	`rule_numbered_paragraph`
`quote_verified`	BOOLEAN		0%	The quote was found verbatim in the source chunk (only verified lessons are served)	`True`
`title`	STRING		0%	Document title (used in citations)	`Cambodia - Health Equity and Quality Improvement…`
`landing_url`	STRING		0%	Document page on documents.worldbank.org (citation link)	`http://documents.worldbank.org/curated/en/099051…`
`doc_date`	STRING		0%	Document date	`2024-05-15`

`gold.rating_evidence` (301 rows)

Grain: one dated DO / IP rating.

Column	Type	Key	Null	Meaning	Example
`project_id`	STRING		0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P131825`
`dimension`	STRING		0%	development_objective | implementation_progress	`development_objective`
`source_type`	STRING		0%	ISR | ICR | ICRR | PPAR	`ISR`
`date`	DATE		0%	Date of the rating	`2019-12-28`
`rating`	STRING		0%	Highly Satisfactory ... Highly Unsatisfactory	`Moderately Satisfactory`
`source_id`	STRING		0%	#ISR etc.: the exact rating record	`33858328#ISR14`
`doc_id`	STRING		0%	Documents & Reports API id (also the PDF file name)	`33858328`
`via`	STRING		0%	Where it was parsed from (e.g. ICR data sheet)	`ICR data sheet`

`gold.project_ratings_resolved` (34 rows)

Grain: project × rating dimension.

Column	Type	Key	Null	Meaning	Example
`project_id`	STRING	PK	0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P131825`
`dimension`	STRING	PK	0%	development_objective | implementation_progress	`development_objective`
`resolved_rating`	STRING		0%	Rating after the evidence hierarchy is applied	`Moderately Satisfactory`
`resolved_from`	STRING		0%	Source type that won (PPAR > ICRR > ICR > latest ISR)	`ICRR`
`resolved_source_id`	STRING		0%	Record that won	`33921463`
`conflict`	BOOLEAN		0%	Sources fall on opposite sides of the satisfactory line	`True`
`spread`	BIGINT		0%	Rating steps between best and worst evidence	`3`
`n_evidence`	BIGINT		0%	Number of rating records	`19`
`worst_isr`	STRING		0%	Worst supervision rating during implementation	`Unsatisfactory`
`trajectory`	STRING		0%	JSON list of (source, date, rating): kept so both sides can be cited	`[["ISR", "2014-08-09", "Satisfactory"], ["ISR", …`

`gold.project_outcomes` (17 rows)

Grain: one completed project.

Column	Type	Key	Null	Meaning	Example
`project_id`	STRING	PK	0%	World Bank project ID (P-code). The conformed key that joins structured and unstructured data	`P152799`
`doc_id`	STRING		0%	Documents & Reports API id (also the PDF file name)	`34382469`
`restructurings`	BIGINT		0%	Number of restructurings (ICR data sheet)	`4`
`closing_planned`	DATE		59%	Original closing date	`2023-12-31`
`closing_actual`	DATE		59%	Actual closing date	`2023-12-31`
`closing_extension_days`	DOUBLE		59%	Days the closing date was extended	`0.0`

Column	Type	Key	Null	Meaning	Example
`outcome_rating`	STRING		6%	Final outcome rating	`Moderately Satisfactory`
`bank_performance`	STRING		6%	Bank performance rating	`Satisfactory`
`me_quality`	STRING		6%	Quality of monitoring and evaluation	`Substantial`
`n_isr_ratings`	BIGINT		0%	ISR ratings found in the data sheet	`17`

`gold.tender_features` (1,632 rows)

Grain: one tender with ≥ invitation date. Inherits every `fact_tender` column; only the added columns are listed.

Column	Type	Key	Null	Meaning	Example
`commitment_usd`	DOUBLE		14%	Total commitment (USD)	`36000000.0`
`approval_date`	DATE		15%	Board approval date	`2020-06-23`
`env_category`	STRING		0%	Environmental category A/B/C/F (older projects)	`B`
`es_risk_rating`	STRING		0%	Environmental & social risk rating (ESF projects; NULL before ESF)	`missing`
`major_sector`	STRING		0%	Major sector of the project	`Education`
`cpia_budget_financial_mgmt`	DOUBLE		45%	CPIA quality of budgetary and financial management (1-6)	`4.0`
`cpia_public_sector_mgmt`	DOUBLE		45%	CPIA public sector management and institutions cluster (1-6)	`3.2`
`cpia_transparency_corruption`	DOUBLE		45%	CPIA transparency, accountability and corruption (1-6)	`3.0`
`wgi_control_of_corruption`	DOUBLE		18%	WGI control of corruption (about -2.5 to 2.5)	`-1.0685743`
`wgi_gov_effectiveness`	DOUBLE		18%	WGI government effectiveness	`-0.6059794`
`wgi_regulatory_quality`	DOUBLE		18%	WGI regulatory quality	`-0.9179055`
`cpia_missing`	DOUBLE		18%	1 when CPIA is not published (non-IDA countries)	`0.0`
`log_contract_usd`	DOUBLE		43%	log(1 + estimated value)	`8.743530038729363`
`log_commitment`	DOUBLE		14%	log(1 + project commitment)	`17.399029524198163`
`project_age_days`	DOUBLE		15%	Days from project approval to invitation	`1801.0`
`open_month`	INT		0%	Month of invitation	`5`
`desc_words`	BIGINT		0%	Words in the description	`12`
`kw_works_civil`	BIGINT		0%	Description mentions works civil keywords (0/1)	`0`
`kw_supply`	BIGINT		0%	Description mentions supply keywords (0/1)	`0`
`kw_consulting`	BIGINT		0%	Description mentions consulting keywords (0/1)	`0`
`kw_it`	BIGINT		0%	Description mentions it keywords (0/1)	`0`
`kw_vehicle`	BIGINT		0%	Description mentions vehicle keywords (0/1)	`0`
`kw_lot`	BIGINT		0%	Description mentions lot keywords (0/1)	`0`
`prior_group_median_days`	DOUBLE		47%	Median cycle of earlier tenders in the same country/category/method (no leakage)	`48.0`
`prior_project_median_days`	DOUBLE		43%	Median cycle of the project's earlier tenders	`31.0`
`prior_project_tenders`	BIGINT		0%	Earlier tenders in the project	`49`

`gold.tender_benchmarks` (67 rows)

Grain: benchmark cohort.

Column	Type	Key	Null	Meaning	Example
`level`	STRING		0%	Fallback level: country+category+method → category+method → category	`country+category+method`
`country_iso3`	STRING		51%	Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)	`BGD`
`procurement_category`	STRING		45%	Works | Goods | Consultant Services | Non-consulting Services	`Goods`
`method_code`	STRING		21%	Method code: RFB, RFQ, QCBS, CQS, INDV, DIR, ...	`RFB`
`n`	BIGINT		0%	Tenders in the cohort (awarded + censored)	`23`
`n_awarded`	BIGINT		0%	Awarded tenders in the cohort	`23`
`median_days`	DOUBLE		1%	Kaplan-Meier median days to award (censoring-aware)	`134.0`
`p80_days`	DOUBLE		6%	Kaplan-Meier 80th percentile days to award (NULL when not reached)	`228.0`
`naive_median_awarded_only`	DOUBLE		0%	Median when still-open tenders are ignored (shows the bias)	`134.0`
`item_type`	STRING		55%	What was bought, classified into the scope's taxonomy (e.g. PCR, imaging, reagents, analyzers)	`Biosafety, sterilization and cold chain`

`gold.kg_edges` (115,327 rows)

Grain: one typed link.

Column	Type	Key	Null	Meaning	Example
`src_type`	STRING		0%	Source entity type	`CONTRACT`
`src_id`	STRING		0%	Source entity id	`1755492`
`rel`	STRING		0%	Relationship	`OF_TYPE`
`dst_type`	STRING		0%	Target entity type	`ITEM_TYPE`
`dst_id`	STRING		0%	Target entity id	`IT, furniture and vehicles`
`method`	STRING		0%	exact_key | fuzzy_desc | fuzzy_name | temporal | llm_extracted | human_confirmed | rule_*	`keyword`
`confidence`	DOUBLE		0%	0-1 confidence of the link	`0.7`

9. Physical design (Delta DDL)

The full DDL for all Gold tables is in `01_DATA_MODEL_DDL.sql`. Example:

```
CREATE TABLE IF NOT EXISTS gold.fact_contract (
  wb_contract_number            STRING NOT NULL COMMENT 'World Bank contract number',
  supplier_id                   STRING COMMENT 'WB supplier id (kept as evidence; not trusted alone)',
  supplier_name                 STRING COMMENT 'Supplier name as published',
  supplier_country              STRING COMMENT 'Supplier country',
  supplier_country_code         STRING COMMENT 'Supplier country ISO-2',
  project_id                    STRING COMMENT 'World Bank project ID (P-code). The conformed key that joins structured and unstructured data',
  project_name                  STRING COMMENT 'Project name',
  country_iso3                  STRING COMMENT 'Country, ISO 3166 alpha-3 (ISO-2 and spelling variants resolved via alias table)',
  region                        STRING COMMENT 'World Bank region of the borrower country',
  fiscal_year                   BIGINT COMMENT 'WB fiscal year of the award',
  signing_date                  DATE COMMENT 'Contract signing date',
  procurement_category          STRING COMMENT 'Works | Goods | Consultant Services | Non-consulting Services',
  procurement_method            STRING COMMENT 'Full method name (Request for Bids, QCBS, ...)',
  review_type                   STRING COMMENT 'Prior | Post review by the Bank (''unknown'' when not published)',
  amount_usd                    DOUBLE COMMENT 'Contract amount (USD)',
  description                   STRING COMMENT 'Free-text description of the package / contract',
  borrower_ref                  STRING COMMENT 'Borrower contract reference as published',
  borrower_ref_key              STRING COMMENT 'Normalised borrower reference (exact-key link to tender_id)',
  supplier_name_key             STRING COMMENT 'Normalised supplier name (legal forms, M/S, punctuation removed)',
  item_type                     STRING COMMENT 'What was bought, classified into the scope''s taxonomy (e.g. PCR, imaging, reagents, analyzers)',
  in_domain                     BOOLEAN COMMENT 'Item type belongs to the domain (laboratory and diagnostic equipment)',
  supplier_key                  STRING NOT NULL COMMENT 'Resolved supplier: normalised name key + supplier country (after fuzzy merge)',
  tender_id                     STRING COMMENT 'Normalised bid reference (upper-case, punctuation removed); the tender''s identity',
  tender_link_method            STRING COMMENT 'How the contract was linked to its tender: exact_key | fuzzy_desc',
  tender_link_confidence        DOUBLE COMMENT 'Confidence of the tender link (1.0 exact; similarity otherwise)',
  state                         STRING COMMENT 'Sub-national state (India scope only)',
  CONSTRAINT pk_fact_contract PRIMARY KEY (wb_contract_number, supplier_key) RELY,
  CONSTRAINT fk_fact_contract_tender_id FOREIGN KEY (tender_id) REFERENCES gold.fact_tender,
  CONSTRAINT fk_fact_contract_project_id FOREIGN KEY (project_id) REFERENCES gold.dim_project,
  CONSTRAINT fk_fact_contract_supplier_key FOREIGN KEY (supplier_key) REFERENCES gold.dim_supplier
)
CLUSTER BY (item_type, country_iso3)
COMMENT 'one contract-supplier row. Source: Major Contract Awards (FinancesOne)'
TBLPROPERTIES (delta.enableChangeDataFeed = true);
```

Scaling. The full public corpus is about 300k contract rows, about 425k notices, about 22k projects and tens of thousands of documents. That is small for Delta. Liquid clustering on country and year keeps scans narrow, Change Data Feed makes Silver → Gold and the vector index incremental, and supplier resolution uses blocking so it stays near-linear.

10. Built on Databricks

The model was created on Databricks on 2026-10-27 by notebooks 00 and 20-34, in catalog `proc_demo`:

Object	Rows on Databricks
`bronze.contract_awards_raw` (Health Goods awards of the domain projects)	14,622
`bronze.domain_projects`	179
`bronze.proc_notices_raw`	15,164
`gold.v_lab_contracts` (lab and diagnostic contract rows)	2,221 (US$1,375,796,408)
`gold.fact_tender` / with an invitation-to-award time	10,407 / 1,068
`gold.dim_supplier`	7,543
`bronze.project_docs_meta` / `gold.proj_doc_chunks`	128 / 6,126
`gold.lessons` (quote-verified)	733 (712)
`gold.kg_edges`	116,199
`gold.rag_chunks` (KEN 1,239, IND 2,556)	3,795